# Genetics, atrophy, liquid biomarkers and cognitive tests in Alzheimer's disease: an integrated and multiomics analysis pipeline to understand the neurodegenerative mechanisms

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## Abstract & Poster description

**Abstract**

- Alzheimer's disease (AD) is the most common neurodegenerative disease among the elderly (50–75%) and the causes of the sporadic form are not yet clear[1]. The genetic predisposition is relevant (60–80%)[2], but alone is not sufficent to trigger those initial processes that lead to neurodegeneration and to the slowly atrophy of the brain.

- An appropriate integration of various types of data (e.g., biomedical images, GWAS/WES, cognitive tests, liquor protein expression) is pivotal to understand this phenomenon[3,4].

**Objectives**

- Build a «Merged AD Dataset»;
- Develop a multikernel or deep learning pipeline to analyze the merged dataset; (public tool for non-data analysis experts;
- Unveil multimodal features (e.g., genes, proteins, metabolites…) able to explain different atrophy levels affecting the most relevant brain regions in the AD.

**Bulletpoints of the work presented in this poster**

- **Presentation of a previous published work** in which we identified **two SNPs signatures** answering two biological questions : AD vs. CN and High vs. low risk of developing AD based on the n° of copies of APOE4 (know risk factor for AD).

- **Presentation of the ongoing work** where we:

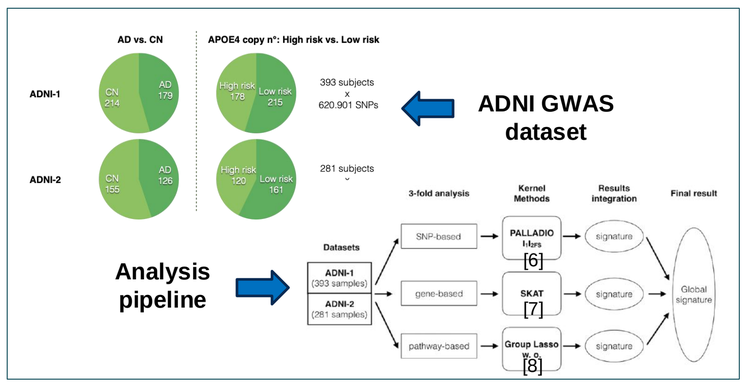
  - validated the SNP signatures of the previous work in the Whole Genome Sequecing (WGS) dataset of ROSMAP (see table of the balanced accuracy);

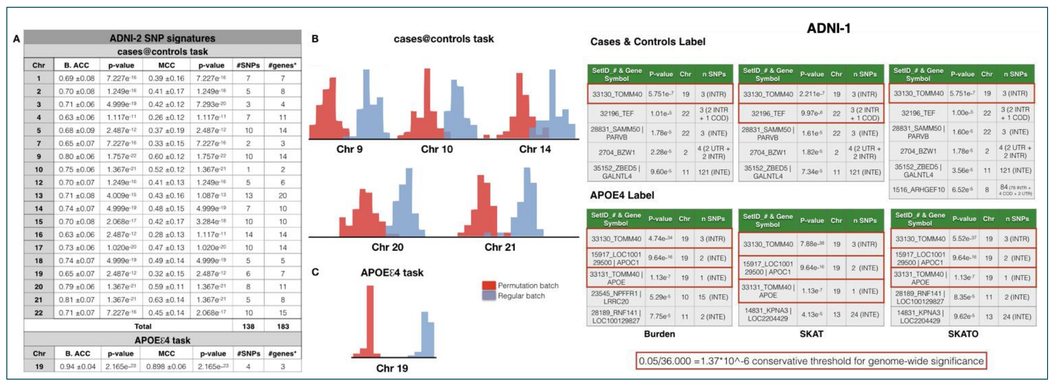
  - studied more deeply the SNP signature related to the higher vs. lower risk of developing AD (see table of the PCA loads);

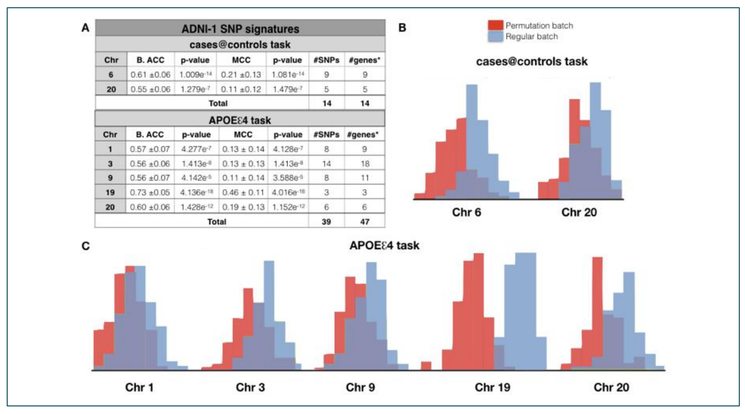
  - performed data integration by MOVICS tool of the RNA-Seq, Metabolome and WGS-filtered ROSMAP dataset;

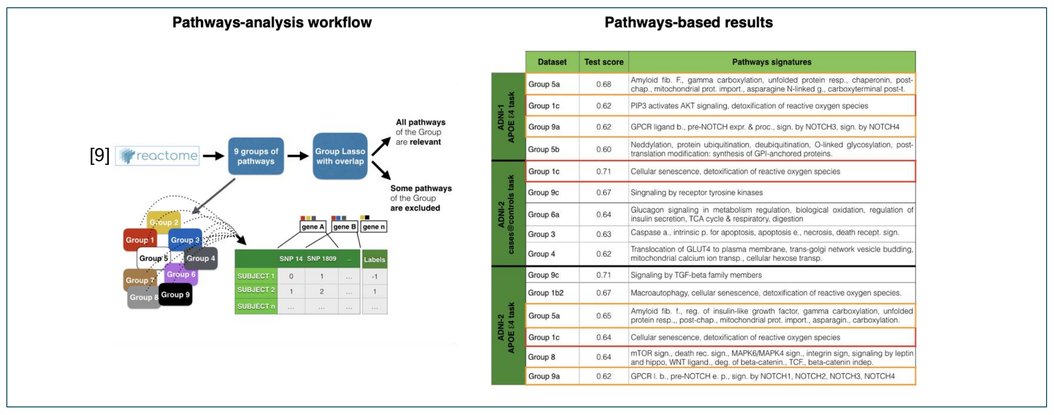
  - analyzed the most discriminanting features (genes and metabolites) among the four clusters identified by MOVICS.

## Previous study on a GWAS dataset of ADNI[5]




### Conclusions

- Possible role of gene **SHLD1** (chr. 20) and its SNPs (rs6053572, rs236137 and rs1287032) in AD. SHLD1 has been **identified in both the classification tasks addressed in ADNI-1 dataset.** SHLD1 is implicated in the DNA repairing mechanism.

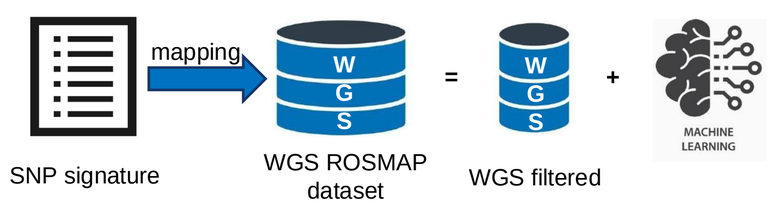
- In ADNI-2**, the identified SNP signature** associated to the AD risk it's very interesting from both the statistical and biological viewpoints: (i) its classification performances are very good (the balanced accuracy is 94% and the MCC is close to 1 (that indicates perfect classification prediction) and (ii) their SNPs lie on a known linkage region associated to AD on chr. 19 (i.e.,19q13.31), but they map close to genes (i.e., LOC101928063, ZNF221 and ZNF155) not associated to AD or to other neurodegenerative disorders.

- "**Neuroactive ligand-receptor interaction**" KEGG pathway was found enriched in both the SNPs signatures identified in ADNI-1 and ADNI-2. In this pathway we found genes that encodes for glutamate receptors. GRM7 (common between ADNI-1 and ADNI-2) plays an important role as neurotransmitter in the cerebral cortex, hippocampus, and cerebellum.

## Data integration in ROSMAP[10] (ongoing work)

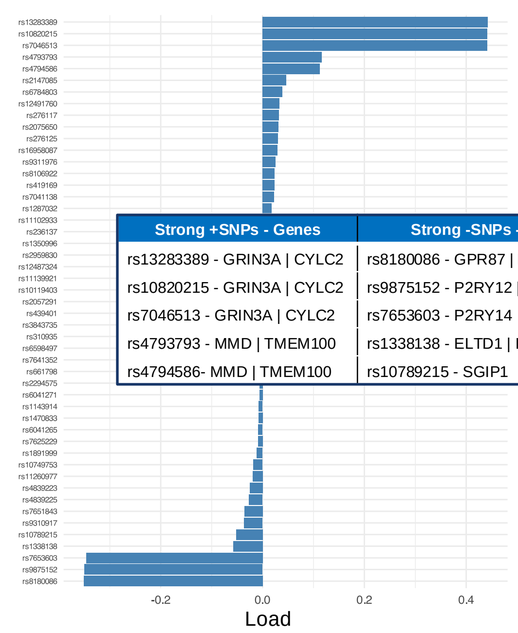
**Integrated data analysis of ROSMAP dataset (preliminary results)**

**Validation of the PALLADIO SNP signature in ROSMAP WGS dataset**



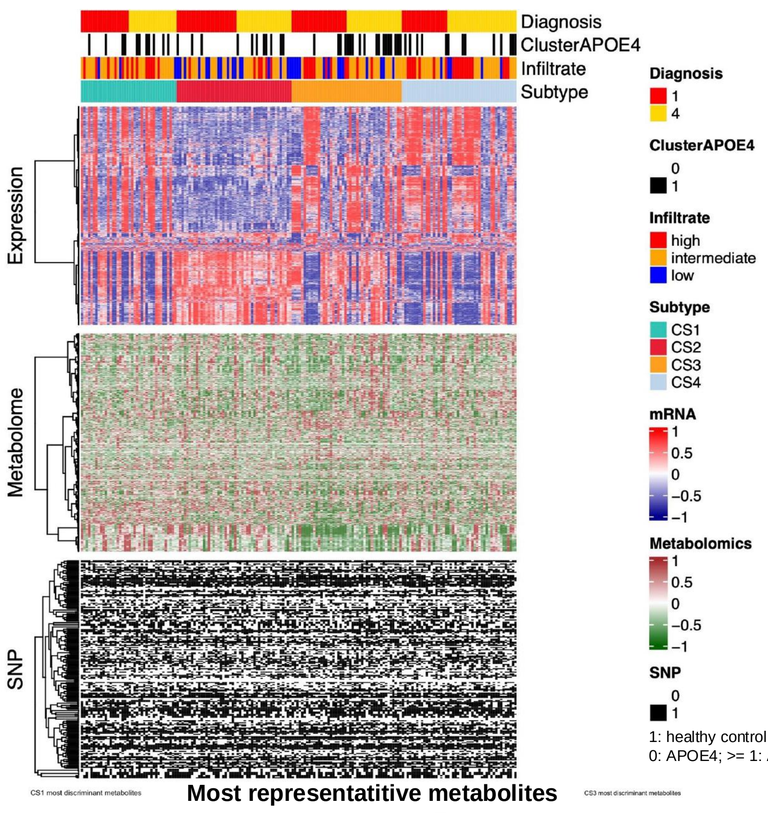

| Biological questions | Balanced accuracy |
|---|---|
| High risk vs Low risk (based on n°APOE4) | 0.80 |
| AD vs CTRL | 0.54 |

**PCA loadings of the SNPs within the high vs. low risk SNP signature**

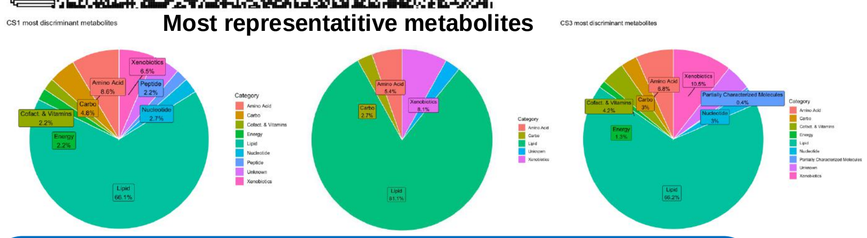



| Strong +SNPs - Genes | Strong -SNPs - Genes |
|---|---|
| rs13283389 - GRIN3A \| CYLC2 | rs8180086 - GPR87 \| P2RY13 |
| rs10820215 - GRIN3A \| CYLC2 | rs9875152 - P2RY12 \| MED12L |
| rs7046513 - GRIN3A \| CYLC2 | rs7653603 - P2RY14 |
| rs4793793 - MMD \| TMEM100 | rs1338138 - ELTD1 \| LOC729817 |
| rs4794586- MMD \| TMEM100 | rs10789215 - SGIP1 |

**Data integration among RNA-Seq, Metabolome and WGS-filtered ROSMAP database with MOVICS[11]**

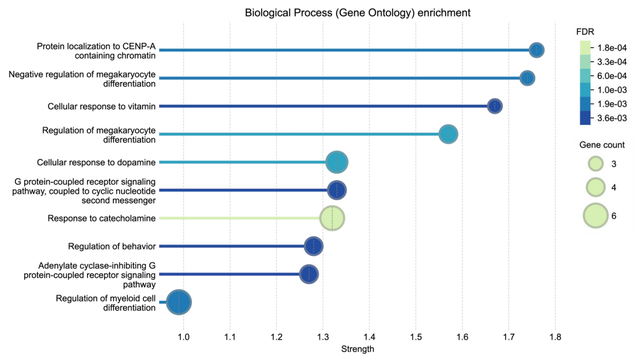



**Most representatitive metabolites**

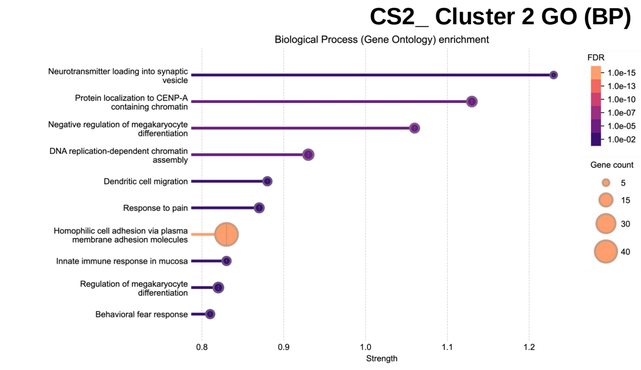


**In silico functional characterization in Gene Ontology[12] (GO)**

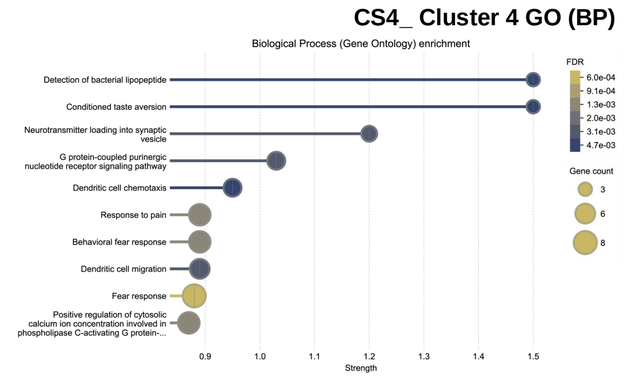
**CS1_ Cluster 1 GO (BP)**




**CS2_ Cluster 2 GO (BP)**




**CS4_ Cluster 4 GO (BP)**




**Perspectives**

- Add other ROSMAP dataset & refine data integration
- Add correlation analysis with clinical but above all with imaging data (e.g., FDG-PET) to identify endophenotypes chacterized by relevant correlations with omics features

## REFERENCES

- [1] Van Cauwenberghe, C *et al*. "The genetic landscape of Alzheimer disease: Clinical implications and perspectives". Genet. Med., 2016.
- [2] Naj, A. C. *et al*. "Genomic variants, genes, and pathways of Alzheimer's disease: An overview". Am. J. Med. Genet. B. Neuropsychiatr. Genet., 2017.
- [3] R. Sims *et* al. "The multiplex model of the genetics of Alzheimer's disease," Nat. Neurosci., 2020.
- [4] Turrisi R, **Squillario M**, et al. **"**An overview of data integration in neuroscience with focus on Alzheimer's Disease**"**. IEEE J Biomed Health Inform. 2023**.**
- [5] **Squillario M** et al. "A telescope GWAS analysis strategy, based on SNPs-genes-pathways ensamble and on multivariate algorithms, to characterize late onset Alzheimer's disease." Sci Rep. 2020**.**
- [6] ADNI: https://adni.loni.usc.edu/
- [7] **PALLADIO $I_3I_{2FS}$**: https://slipguru.github.io/palladio/framework.html
- [8] **SKAT:** https://github.com/leelabsg/skat
- [9] REACTOME: Jassal B et al. "The reactome pathway knowledgebase". Nucleic Acids Res. 2020 Jan 8.
- [10] ROSMAP: https://adknowledgeportal.synapse.org/Explore/Studies/DetailsPage/StudyDetails?Study=syn3219045
- [11] **MOVICS**: Lu X et al**. "**MOVICS: an R package for multi-omics integration and visualization in cancer subtyping". Bioinformatics. 2021 Apr 1.
- [12] Ashburner M et al. "Gene ontology: tool for the unification of biology. The Gene Ontology Consortium. ". Nat Genet. 2000 May

**ROSMAP data**

WGS data: doi:10.1038/mp.2017.20.

RNAseq: doi:10.1038/s41593-018-0154-9

HD4 Metabolomics: : doi:10.3233/JAD-179939

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